

Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations

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Abstract

Objectives: Model specification—what adjusting variables are analytically modeled—may influence results of observational associations. We present a standardized approach to quantify the variability of results obtained with choices of adjustments called the “vibration of effects” (VoE).

Study Design and Setting: We estimated the VoE for 417 clinical, environmental, and physiological variables in association with all-cause mortality using National Health and Nutrition Examination Survey data. We selected 13 variables as adjustment covariates and computed 8,192 Cox models for each of 417 variables’ associations with all-cause mortality.

Results: We present the VoE by assessing the variance of the effect size and in the $-\log_{10}(P\text{-value})$ obtained by different combinations of adjustments. We present whether there are multimodality patterns in effect sizes and P -values and the trajectory of results with increasing adjustments. For 31% of the 417 variables, we observed a Janus effect, with the effect being in opposite direction in the 99th versus the 1st percentile of analyses. For example, the vitamin E variant α -tocopherol had a VoE that indicated higher and lower risk for mortality.

Conclusion: Estimating VoE offers empirical estimates of associations are under different model specifications. When VoE is large, claims for observational associations should be very cautious. © 2015 Elsevier Inc. All rights reserved.

Keywords: Vibration of effects; Environment-wide association study; Model specification; Biostatistics; Observational association; Confounding

1. Introduction

Observational associations between variables do not guarantee causality, and they are often complex and influenced by other variables (confounders and effect modifiers). Accounting for covariates is typically achieved through statistical modeling, such as multivariate regression. However, what variables should one choose to account for in complex multivariate phenomena where many variables may be confounded or correlated [1]? *Model specification* can be a major issue in diverse fields,

including epidemiology [2], economics [3–5], and psychological science and neurosciences [6]. Thousands of associations are published, and many are often challenged and refuted by subsequent investigations [7–9]. Choices of models underlie our assumptions about association and about potential causes and effect [10]. Very often there is large uncertainty about what variables should be modeled and how they are related. Consequently, there is large heterogeneity in how investigators associate variables [2].

In discovery-based research in large data sets, there is often no prior evidence or biological plausibility on what adjustment variables to include in statistical models. In other cases, unequivocal evidence and plausibility may exist to include some adjustment variables in the model, lack of consensus on some others, and no available guidance on yet another set of adjustment variables. Interpretation of effects may vary depending on the analytical choices made. A way to compute the extent of instability

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What is new?**Key findings**

- We have developed a standardized, automated procedure to estimate the “vibration of effects” (VoE) to measure the instability of observational associations between variables under different modeling scenarios; for example, the vitamin E variant α -tocopherol had a VoE that indicated both higher and lower risk for mortality.

What this adds to what was known?

- Model selection influences how we infer from associations, such as hazard ratios and *P*-values. In this report, we provide a way to measure and visualize how much model selection influences variability in hazard ratios and *P*-values.

What is the implication and what should change now?

- The VoE can offer a proper picture of the instability of postulated associations and allow more judicious claims for associations in observational studies; VoE can be widely adopted in observational studies.

of the results due to model specification is needed to guide inference.

The “vibration of effects” (VoE) [2] describes the extent to which an estimated association changes under multiple distinct analytical modeling approaches. The VoE is related also to the previously described concept of “multiple modeling” [9] or statistical model-induced variability [11]. To estimate the VoE empirically, we can compute the distribution of the point estimates of measures of association (e.g., relative risks, odds ratios) and *P*-values that are possible under different analytical scenarios. The VoE measures how susceptible an association is under different modeling scenarios; the larger the VoE, the greater the instability of the results. One may also explore which specific scenarios most influence the estimated association. Here, we describe a framework to systematically evaluate the VoE for a set of adjustment covariates.

2. Example of a controversial association

As an introductory example, we use the VoE framework to evaluate a contentious association between vitamin E (α -tocopherol) and mortality. Early publications of observational studies claimed large reductions in disease-related and mortality-related events in association with vitamin E [12,13]. However, clinical trials that followed were not able

to support the early observational findings [14–17]. Furthermore, meta-analyses of clinical trials have showed nearly the opposite of early observational studies including null [18] to even *increased* risk [19,20] of vitamin E on adverse health-related outcomes including mortality. An important question is to understand the extent to which the results of observational studies on vitamin E may depend on how the observational data are analyzed and in particular, on the model specification, that is, which other factors are taken into account in multivariate modeling. For a resolution, see the VoE analysis for vitamin E and mortality at the end of the Section 4.

3. Methods

3.1. Data source: NHANES 1999–2000, 2001–2002, and 2003–2004

We downloaded National Health and Nutrition Examination Survey (NHANES) examination, laboratory, questionnaire, and National Death Index (NDI) linked mortality data for 1999–2000, 2001–2002, and 2003–2004 surveys. Mortality information was collected from the date of the survey participation through December 31, 2006, and ascertained via a probabilistic match between NHANES and NDI death certificate information. The NDI matches individuals on personal and demographic criteria, such as social security number and date of birth and its performance has been described elsewhere [21]. Overall, 9,555, 11,021, and 10,100 participants were followed in the 1999–2000, 2001–2002, and 2003–2004 surveys, respectively, with 611, 470, and 276 assumed death events.

3.2. Variables of interest

A total of 993 self-reported, clinical, and molecular phenotypes were assayed on participants of NHANES between 1999 and 2004, and we associated 417 of these with all-cause mortality that had a minimum sample size of 1,000 and a minimum of 100 deaths. These variables included (1) 179 serum or urine biomarkers of environmental exposures; (2) 9 self-reported behavioral factors such as smoking, alcohol consumption, and physical activity; (3) 85 self-reported nutritional intake information; (4) 27 self-reported health conditions; (5) 92 clinical factors such as body mass index (BMI), blood pressure, and cholesterol; and (6) 13 demographic variables.

Environmental exposure variables are physical/chemical biomarkers of external exposures measured in serum or urine, such as blood lead concentration or assays of infectious agents, such as human immunodeficiency virus. Broadly, these included a serum marker of nicotine metabolism (cotinine), dioxins ($n = 7$ markers), furans ($n = 10$), heavy metals ($n = 15$), hydrocarbons ($n = 21$), nutrients ($n = 15$), polychlorinated biphenyls ($n = 34$), pesticides ($n = 22$), phthalates ($n = 12$), estrogenic

compounds ($n = 6$), bacterial ($n = 2$), and viral organisms ($n = 6$; see Table S2/Appendix at www.jclinepi.com). With exception of assays detecting infectious agents (which were positive/negative assays), factors were continuous in scale. All continuous exposure markers were mean subtracted and standardized by the standard deviation (z standardized) to facilitate comparisons.

Self-reported behavioral factors included current/past smoker (versus never smoker), alcohol consumption (e.g., “have you ever had five or more drinks per day?” “how many drinks per day in last month?”), and physical activity. Physical activity was estimated by deriving metabolic equivalents for self-reported leisure and normal-time activities [22] and treated as ordinal variables based on Health.gov physical activity guideline categories for no aerobic activity, low activity (medium intensity activity greater than baseline but fewer than 150 min/wk), moderate activity (150–300 medium intensity minutes a week), and high activity (>300 minutes of medium intensive activity per week or >150 minutes of high intensity per week) as previously described [23,24].

Self-reported food and nutrient consumption variables were determined from one in-person 24-hour interview (1999–2000, 2001–2002) or two 24-hour (2003–2004) in-person and on the phone interviews using the US Department of Agriculture and Department of Health and Human Services food recall questionnaires [25–28]. These included daily estimated intake of 68 nutrients and vitamins, including vitamins A, B, C, and E; carotenoids; carbohydrates; sugars; proteins; fats; minerals (iron, sodium, potassium). All continuous food and nutrient consumption variables were z standardized for comparison.

Clinical variables included body measurements assessed by a health professional, including height, weight, waist circumference, thigh circumference, triceps skinfold, subscapular skinfold, calf circumference. Laboratory values included serum measures of lipids (e.g., fasting low-density lipoprotein cholesterol, high-density lipoprotein [HDL] cholesterol, triglycerides, glucose, insulin), creatinine (urine and serum), albumin (serum and urine), uric acid, total serum protein, C-reactive protein, thyroxine, and thyroid-stimulating hormone, and others. Laboratory measures were log transformed. All continuous clinical measures were z standardized to facilitate comparison.

3.3. Adjusting variables

We chose a set of 15 variables as our set of possible adjustments. Because there is no perfect consensus on what variables should (or should not) be included as adjustments in association to all-cause mortality, we based our set on a large meta-analysis of 80 studies of physical activity on all-cause mortality [29]. Because age and sex are well-known factors related to mortality, we chose to keep these in all models (“baseline” variables) and then chose 13 other

variables as the set of varying adjustments, the 12 that had the highest frequency of use among these 80 investigations (smoking, BMI, hypertension, diabetes, cholesterol, alcohol consumption, education, income, family history of heart disease, heart disease, any cancer, physical activity) and race/ethnicity, because NHANES samples are racially heterogeneous previous studies examining mortality in NHANES adjusted for race/ethnicity [30,31].

We coded the 15 adjustment variables as binary, categorical, or continuous variables. Medical history items, including any hypertension, diabetes, heart disease, family history of heart disease, and cancer were treated as binary variables. We treated alcohol consumption as a binary variable (drinking five drinks per day: yes/no). Physical activity was estimated by deriving metabolic equivalents for self-reported leisure and normal-time activities [22] and treated as ordinal variables based on Health.gov physical activity guideline categories as previously described [23]. Serum total cholesterol in NHANES was assayed in participants who had fasted at least 8 hours before the survey. Smoking was coded as a three-category variable (self-reported current or past smoker vs. never smoker). BMI was coded as five categories based on recent investigations between BMI and all-cause mortality [32], including underweight (BMI <18.5 kg/m²), overweight (BMI ≥25 and <30 kg/m²), obese (BMI ≥30 kg/m² and <35 kg/m²), or very obese (BMI ≥35 kg/m²) with the reference group being normal weight (BMI ≥18.5 and <25 kg/m²). Income was coded as tertile of income to poverty ratio (first and second tertile vs. third tertile). Education was coded as less than high school education or high school equivalent (vs. greater than high school education). We added race/ethnicity as NHANES participants are representative of general population of the United States (e.g., whites, blacks, and Hispanics), and previous studies examining mortality in NHANES have routinely adjusted for race/ethnicity [30,31]. Race/ethnicity was coded as Mexican American, non-Hispanic black, other Hispanic, and other (vs. non-Hispanic whites). We describe the adjustment variables for survivors and deceased individuals in Table S1/Appendix at www.jclinepi.com.

For analyses of the association of serum Vitamin D with all-cause mortality, we increased the number of rotating adjusting covariates to 19 (from 13) to observe how the VoE changes under different sets of adjusting covariates, as the options increase for adjustment further. In addition to the 15 variables mentioned previously, we selected urinary creatinine, urinary albumin, and serum calcium as documented in a recent investigation of vitamin D levels in mortality [33]. Furthermore, we chose serum cadmium as we recently documented a potential association with mortality [34] along with two serum nutrient indicators, β carotene and γ tocopherol, which were associated with type 2 diabetes and cholesterol lipid levels [23,35] and had consistent VoE patterns.

3.4. Assessing the vibration of effects in all-cause mortality

We modeled 417 variables separately as a function of time to death with the Cox proportional hazards model. VoE may be used with any linear model, including linear or logistic regression. We used the *R* project *survival* and *survey* library for all analyses and accounted for clusters pseudostrata, pseudosampling units, and participant weights to accommodate the complex sampling of the data [36,37].

We considered the VoE in the context of 13 adjustment variables not including age and sex. The total number of possible combinations of adjusting variables from the set of 13 total adjustments is 8,192. For all 8,192 models, sample sizes remained constant. Our algorithm for computing the VoE for a variable x (e.g., HDL cholesterol) is the following:

- (1) Select a number k from 0 to n total adjustment variables ($n = 13$). Categorical variables are considered as a single group (e.g., education or BMI is selected as a group of variables together)
- (2) For each combination of k number of adjustment variables (k number of distinct variables from the set of all variables z_0 through z_n), associate x with all-cause mortality with Cox proportional hazards and assess the effect size (β) and P -value for x :

$$\text{Hazard Ratio} = \exp(\beta x + \gamma_1 \text{age} + \gamma_2 \text{male} + i_0 \delta_0 z_0 + \dots + i_n \delta_n z_n)$$

Where $i_0 \dots i_n$ are indicator variables ($=1$ or 0) whether the particular variable is in the model or not for a given k . Thus, the sum of indicator variables is equal to k . The number of iterations in this step is the total number of k number of combinations possible out of a set of n total variables.

- (3) If $k = n$, then stop. If $k < n$, then go back to step 1. for $k+1$

3.5. Assessing summary statistics

We computed summary statistics for the VoE. These included the 1st, 50th (median), and 99th percentile of effect sizes and P -values. We created summary statistics derived from the distributions of VoE for a variable. The first was the “relative hazard ratio” (RHR), the ratio of the 99th percentile and 1st percentile HR (or the difference between the 99th and 1st percentile of effect size estimates). The RHR measures the difference between the larger and smaller effect sizes and this measure connotes a spread of effect sizes for different combinations of adjustments. The second was the “relative P -value” (RP), which is the difference between the 99th and 1st percentile of $-\log_{10}(P\text{-value})$. We assessed the median effect size and P -value for a given variable for each k th level of adjustment.

3.6. Systematic search for interactions between variables and adjustments

The VoE may vary depending on different strata of adjustment variables. Thus, we systematically searched for interactions between variables of interest and the 13 adjustment variables with the inclusion of a multiplicative interaction term. Interactions between variables are very often claimed in observational research, but the search for them is not systematic and the validity of the claimed interactions is often tenuous [38]. Specifically, our procedure to search for interactions between a variable x and the 13 possible adjusting variables was the following:

For each adjusting variable k from 1 to 13:

$$\text{Hazard Ratio} = \exp(\beta x + \gamma_1 \text{age} + \gamma_2 \text{male} + \delta_k z_k + \zeta_k x z_k)$$

Where ζ_k corresponds to the interaction term. We collected the test statistic and P -value for the interaction term to determine the presence of interactions. For categorical adjustment variables, we assessed presence of interaction each category of the variable separately. Thus, we tested for interactions between variables of interest and two categorical variables corresponding to education (less than high school and high school education vs. greater than high school education), two variables corresponding to the largest race groups (non-Hispanic whites and Mexican Americans vs. whites), four BMI variables (vs. normal weight), two variables corresponding to current and past smoking, and one variable each for coronary disease, family history for coronary disease, hypertension, diabetes, any cancer, total cholesterol (mean subtracted standardized by the SD), physical activity, drink five drinks per day, age, and male sex resulting in 20 effective tests of interaction for a variable x . To maximize power of detection of interaction, we restricted the search to 107 variables with the least missing data and sample sizes $>6,500$ (range of sample size, 6,500–8,607). Thus, this resulted in 2,140 total of interactions tested ($107 \times 22 = 2,140$). We considered the strongest interactions as those that surpassed a Bonferroni level of statistical significance ($P < 0.05/2,140 = 2 \times 10^{-5}$).

For interactions that surpassed Bonferroni level of significance, we computed the VoE for each strata of the adjustment variable separately. For example, if we found a significant interaction between a variable x and any family history of heart disease, we assessed the VoE for x for samples who answered without family history of heart disease and with family of heart disease separately. If an interaction was found between a variable x and total cholesterol, we categorized samples as having <200 mg/dL, ≥ 200 and <239 mg/dL, and >239 mg/dL total cholesterol. We show the VoE for the strongest interaction between platelet count and total cholesterol (interaction $p = 1 \times 10^{-10}$, total cholesterol main effect $P = 0.01$, and platelet count main effect $p = 1 \times 10^{-9}$) in Fig. S4/Appendix at www.jclinepi.com.

4. Results

4.1. Estimating the VoE

VoE is estimated by computing the hazard ratio (HR) and *P*-value for a variable of interest while adjusting for all possible combinations of adjustments from a finite set of adjustment variables. Our algorithm for computing the VoE for a variable *x* (e.g., serum vitamin D) is shown in Fig. 1.

First, we downloaded 417 self-reported, clinical, and molecular measures with linked all-cause mortality information in participants from NHANES 1999–2004 (Fig. 1A). Mortality information was collected from the date of the survey participation through December 31, 2006, and ascertained via a probabilistic match between NHANES and National Death Index (NDI) death certificate information [21]. We chose variables of interest that had data on at least 1,000 participants and at least 100 death events during follow-up.

Next, we describe the VoE methodology for the association between serum vitamin D and all-cause mortality (Fig. 1B). The total number of combinations of adjusting variables from the set of $n = 13$ total adjustments is 8,192 (or, in general, 2^n models, Fig. 1C). We chose a set of 13 variables as the set of possible adjustments (Fig. 1B, C, Appendix at www.jclinepi.com). Because there is no consensus on what variables should (or should not) be included as adjustments in association with all-cause mortality, we based the selection of these 13 variables on a large meta-analysis of 80 studies of physical activity on all-cause mortality [29]. The most common adjustment variables in these 80 investigations included (in decreasing order of frequency) age, smoking, BMI, hypertension, diabetes, cholesterol, alcohol consumption, education, income, sex, family history of heart disease, heart disease, and any cancer. Because age and sex are well-known factors related to mortality, we chose to keep these in all models (“baseline” variables). The HR and the respective *P*-value for the association of that variable with all-cause mortality are estimated for all 8,193 models with different combinations of 13 adjusting variables using Cox proportional hazards time-to-event regression (Fig. 1D). We visualized the VoE for a given variable by plotting the HR vs. $-\log_{10}(P\text{-value})$ as two-dimensional histogram and a contour plot (Fig. 1E).

We created metrics to express the distributions of VoE for a variable (Fig. 1F). The first was the RHR, the ratio of the 99th percentile and 1st percentile HR. The RHR connotes the spread of HRs for different combinations of adjustments. The second was the RP, which is the difference between the 99th and 1st percentile of $-\log_{10}(P\text{-value})$. The RP measures the range of *P*-values over all estimates. We also assessed whether associations appeared on both sides of the null ($HR < 1$ and $HR > 1$): depending on what adjustments are chosen, the results may suggest that the variable of interest is associated with either increased or decreased mortality.

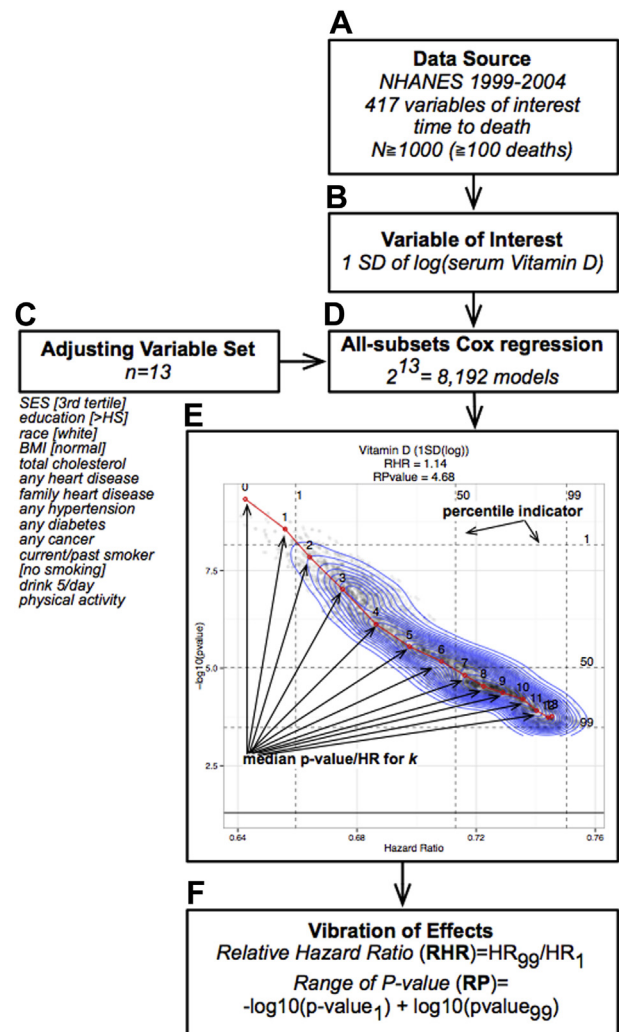


Fig. 1. Vibration of effects (VoE) computation schematic. (A) Data source. (B) Choose a variable of interest. (C) Construct a set of adjustment variables from a set of 13 socioeconomic, demographic, or health-related variables. Reference level is in the square brackets. (D) All subsets Cox regression for each 8,193 models. (E) Visualization (“volcano plot”) of $-\log_{10}(P\text{-value})$ vs. effect size (HR). The median HR and *P*-value of the number of adjustment variables (*k*) in the model is in red. The 1st, median, and 99th percentile of the $-\log_{10}(P\text{-value})$ and HR are depicted in the dotted line. (F) Compute VoE summary statistics, the relative hazard ratio (RHR) and relative *P*-value (RP). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

We also visualized trends corresponding to the number of adjusting variables (*k*), plotting the median effect size and *P*-value for each *k* from 0 to 13. We recorded the proportion of estimates that achieved different levels of nominal statistical significance ($P < 0.05$, 0.0001).

The 417 variables included 179 serum or urine biomarkers of environmental exposures (e.g., serum cadmium, mercury, or pesticide level), 9 self-reported behavioral factors such as smoking and physical activity, 84 self-reported nutritional intake information (from a food frequency questionnaire), 27 self-reported health conditions (e.g.,

diabetes), 92 clinical factors (e.g., BMI and cholesterol), and 13 sociodemographic variables (e.g., income). All continuous variables were log transformed and z standardized for comparison. Appendix (at www.jclinepi.com) describes these 417 variables.

4.2. Prototypical patterns of the VoE

We describe four prototypical patterns from the set of 417 variables (Fig. 2, see Appendix at www.jclinepi.com for all 417 variables). The first pattern is exemplified by the association between serum levels of vitamin D and mortality (Fig. 2A). All the HR estimates are <1.00 , indicating that higher levels of vitamin D tend to be associated with longer survival (all HR <0.76); however, the magnitude of the estimated effect is dependent on the number of adjustment variables, and the association is attenuated when adjusting for more variables, from HR = 0.64 with no adjustment ($k = 0$) to 0.75 with all 13 adjustment variables included ($k = 13$). In contrast, the P -values are less than the nominal level of statistical significance ($P = 0.05$, black line). Most of the results are centered on HR ~ 0.72 and $P \sim 10^{-4}$ (two-dimensional mode). In this first pattern, one concludes that although adjustment weakens the magnitude relationship between vitamin D levels and mortality, inferences regarding the relationship are similar throughout all scenarios of adjustment. Of the 417 variables, 53 (13%) exhibited similar behavior to vitamin D, where all associations were beyond the level of nominal statistical significance, but the association was attenuated with a greater number of adjustment variables (see Fig. S1/Appendix at www.jclinepi.com).

The second pattern is exemplified by the relationship between thyroxine and mortality, displays how increasing adjustment might change inference (Fig. 2B). Higher thyroxine levels tend to be associated with longer survival, but P -values become greater than the nominal level of statistical significance ($P = 0.05$) with nine or more adjustment variables on average. Of the 417 variables, 91 (22%) variables had similar behavior to thyroxine in which HR were attenuated and the P -values rose above the nominal level of significance ($P > 0.05$) as the number of adjusting variables, k , increased (see Fig. S1 and Table S3/Appendix at www.jclinepi.com).

The third pattern, as exemplified by an indicator of kidney function, urinary creatinine, and mortality, shows an opposite trend (Fig. 2C). For $k = 5$ –13 number of adjustment variables, the association tends to become stronger in HR and statistical significance; however, the trend is less clear for $k = 0$ –4, where the median P -values increase. Twenty-six (6%) of the 417 variables exhibited similar behavior to urinary creatinine where the effect sizes increased and P -values decreased for larger k .

In the last pattern, as exemplified by α -tocopherol (Fig. 2D), the estimated HRs can be both greater and less than the null value (HR > 1 and HR ≤ 1) depending on

what adjustments were made. We call this the *Janus effect* after the two-headed representation of the ancient Roman god. For α -tocopherol, most of the HR and P -values were concentrated around 1 and nonsignificance, respectively. However, 1% of the models had an HR < 0.875 (12.5% decreased risk of death for 1SD increase in exposure) with a nominally significant P -value ($P \leq 0.05$), whereas another 1% of the models had HR > 1.05 (5% increased risk for death for 1SD increase of exposure), albeit without reaching nominal significance. The *Janus effect* is common: 131 (31%) of the 417 variables had their 99th percentile HR > 1 and their 1st percentile HR < 1 .

Examples such as those in Fig. 2A–D represented the VoE patterns for 72% of the 417 associations. Other patterns included VoE where all P -values were >0.05 and the strength of the association decreased ($n = 50$, 12%), increased ($n = 27$, 6%), or showed no dependence ($n = 15$, 4%) with increasing number of adjustment variables k (see Table S3/Appendix at www.jclinepi.com). Rarer patterns included variables where all P -values were <0.05 and there was an increasing strength of association ($n = 5$, 1%) or no clear relationship with increasing k ($n = 4$, 1%), and those having P -values with a range less than and greater than 0.05 with no clear relationship with k ($n = 15$, 4%).

4.3. Identifying “multimodality of effects” with VoE

By empirically estimating the VoE, it is also possible to detect whether one or more adjustment variables make a marked difference in the results, leading to multiple modes (Fig. 3) which we call *multimodality of effects*. Multimodality of effects was clearly seen in 71 of the 417 (17%) assessed variables. For example, the overall VoE for serum cadmium on mortality indicates strong association with mortality (Fig. 3A); all of the HRs are >1.2 per 1 SD change in serum cadmium levels, and P -values in all analytical scenarios are <0.05 . However, two modes are visually evident (Fig. 3A). To identify the key variable(s) that separated these different distributions, we visualized the VoE by coloring each point on whether it included (or did not include) each one of the 13 adjustment variables in the model, leading to 13 separate visualizations. In serum cadmium, we observed the two distinct modes were indicative of models that contained or did not contain current or past smoking (Fig. 3B). Specifically, models that contained the smoking adjustment variable (Fig. 3B, yellow points) had HR lower than the models without the smoking adjustment and lower $-\log_{10}(P\text{-values})$ (Fig. 3B, black points). One source of cadmium exposure includes smoking, and we concluded that the correlation between smoking and exposure to cadmium might be driving the multimodal behavior of VoE. Furthermore, we observed that models that included (or did not include) alcohol drinking also resulted in separate modes in P -values (Fig. 3C).

We observed three modes in the association between triglyceride levels and mortality (Fig. 3D–F). The

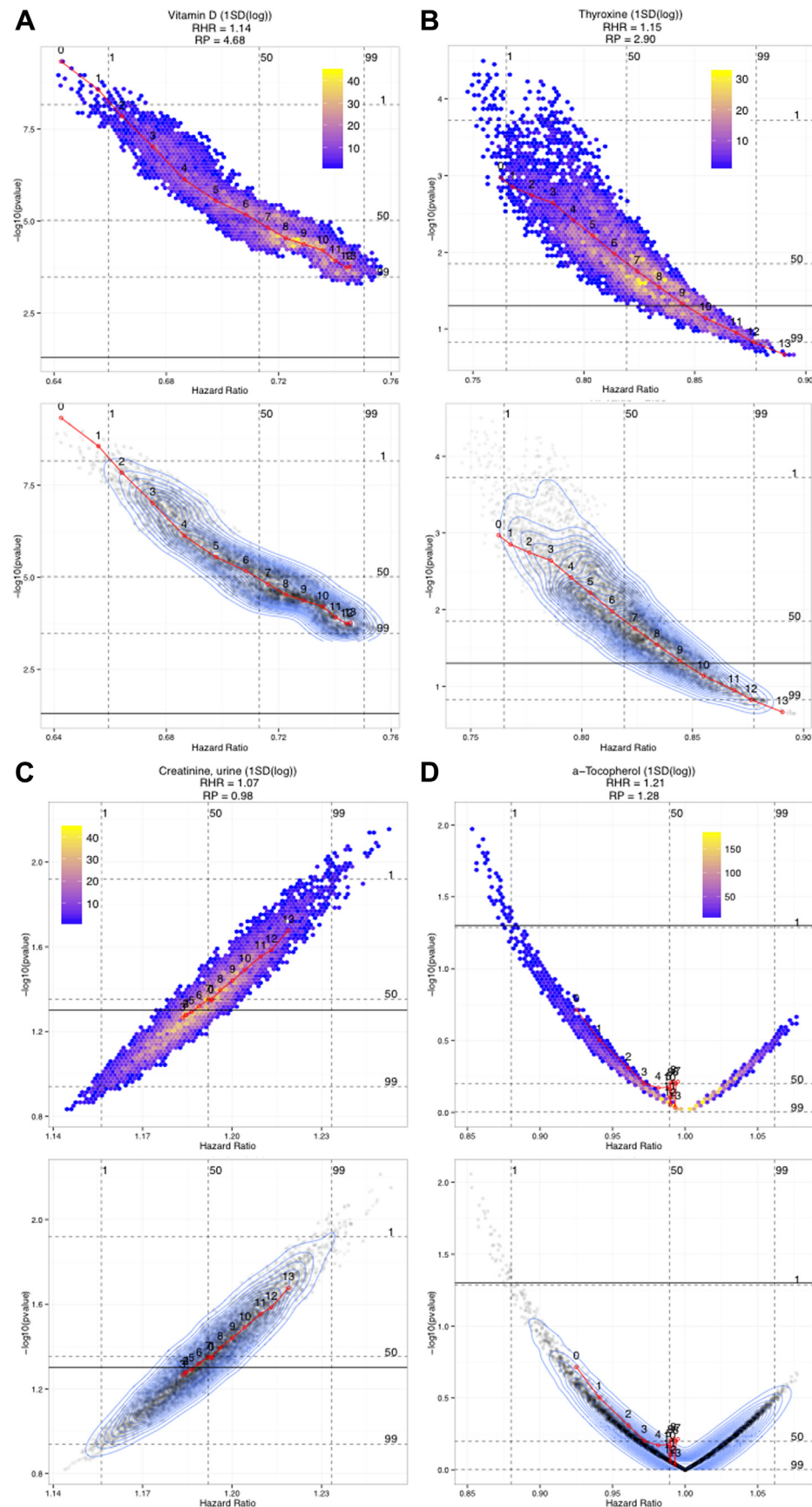


Fig. 2. Volcano plots visualizing the vibration of effects (VoE) for four examples, (A) serum vitamin D, (B) serum thyroxine, (C) urinary creatinine, (D) serum α -tocopherol. Two-dimensional histogram representation in *upper panel* and contour scatter plot is in *lower panel*. All effects are for a 1SD change in logged level of variable interest.

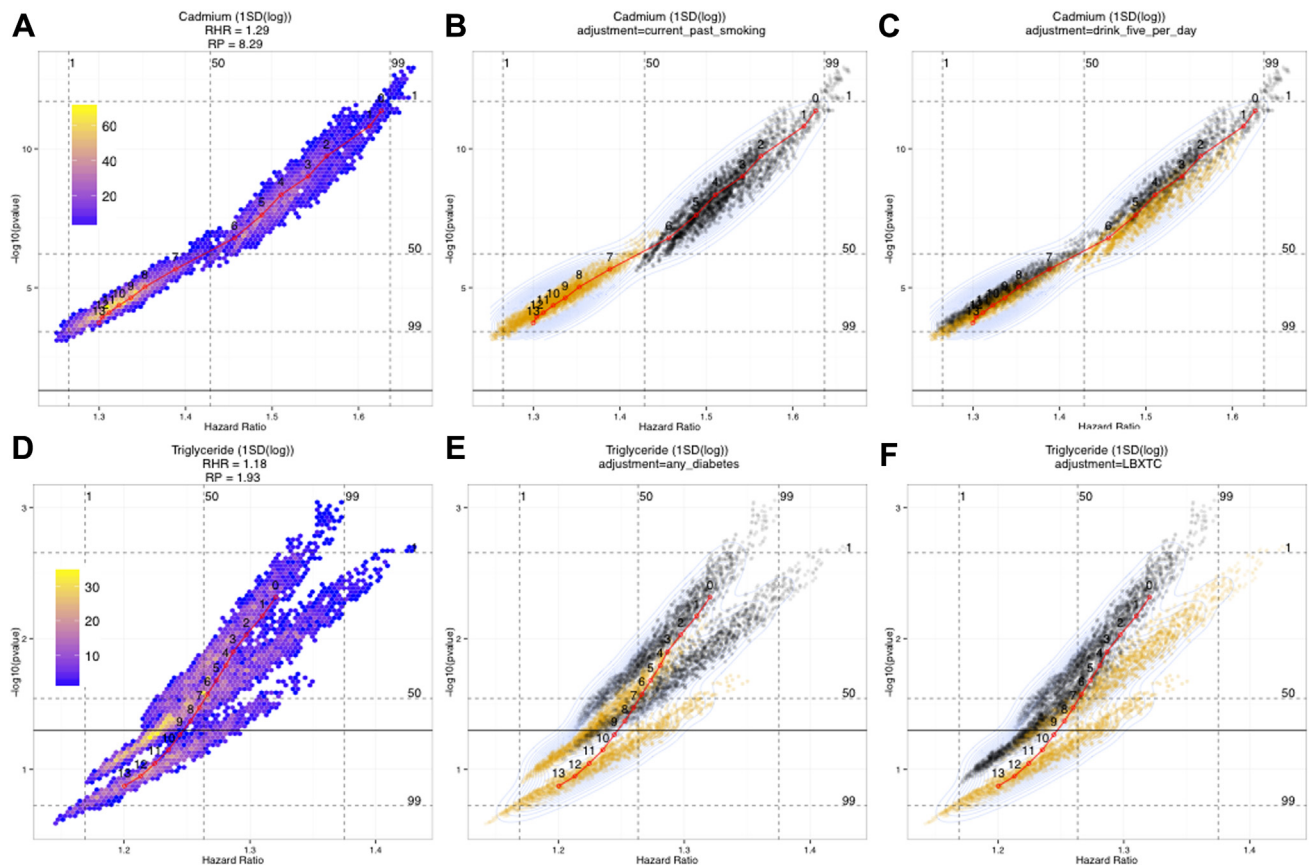


Fig. 3. Volcano plots visualizing vibration of effects (VoE) for three examples with multiple “modes.” (A) The 2D histogram for 1SD increase of the logarithm of serum cadmium, (B) volcano scatter plot with of serum cadmium if smoking was included in the model (yellow) or smoking not included in model (black). (C) Volcano scatter plot for serum cadmium models with drink five per day (yellow) or models without drink five per day (black). (D) The 2D histogram for 1SD increase of the logarithm of serum triglycerides, (E) with total cholesterol included in the model (yellow) or total cholesterol not included in model (black). (F) With any diabetes (yellow) or models without any diabetes (black). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

multimodal plots indicated that total cholesterol and diabetes were driving these modes. For example, in models that did not contain these two adjustments, the associations had smaller P -values and a smaller range of HR. Furthermore, in models containing diabetes, HR were attenuated. The multimodal pattern seems reasonable in light of the high correlation between triglyceride levels and total cholesterol levels/risk for diabetes. We observed a similar pattern for other cardiometabolic indicators, including fasting blood glucose and insulin (see Fig. S1/Appendix at www.jclinepi.com).

4.4. Summary of common patterns of the VoE

Figure 4 shows the distribution of the fold deviation of HR from the null ($HR = 1.00$), the $-\log_{10}(P\text{-value})$, RHR, and RP for all 417 variables considered. The “fold deviation” is the difference of the median VoE-estimated HR from 1 (the null value). The median fold deviation was 1.13-fold (25th percentile: 1.05-fold, 75th percentile: 1.24-fold, Fig. 4A). Moreover, 50% of the variables had a median P -value less than or greater than 0.27 (25th percentile:

0.04, 75th percentile: 0.59, Fig. 4B). The median RHR was 1.15 (5th percentile: 1.07, 25th percentile: 1.11, 75th percentile: 1.22, 95th percentile: 1.70, Fig. 4C). The median RP was 1.07 (5th percentile: 0.31, 25th percentile: 0.589, 75th percentile: 2.03, 95th percentile: 5.09). We observed that most associations could vary by at least 1.15-fold in the magnitude of the HR and by one order of magnitude [$\log_{10}(P\text{-value})$] in the level of statistical significance, and much larger changes were not uncommon. We observed a weak correlation between RHR and RP (see Fig. S2/Appendix at www.jclinepi.com, $\rho = 0.09$, $P = 0.06$).

Returning to the prototypical examples that we discussed previously, the RHR for vitamin D and thyroxine was moderate 1.14 (44th percentile) and 1.15 (51st percentile; Figs. 4C, 2A, and B). However, their RPs were among the largest and equal to 4.7 (93rd percentile) and 2.90 (84th percentile), respectively (Figs. 4D, 2A, and B). For urinary creatinine, the scenarios of adjustment had less prominent VoE. The RHR and RP for urinary creatinine was 1.07 (5th percentile) and 0.98 (47th percentile; Fig. 4C and D). The RHR for α -tocopherol (with the *Janus effect*) was higher (1.21, 71st percentile, Fig. 4C). Variables that demonstrated

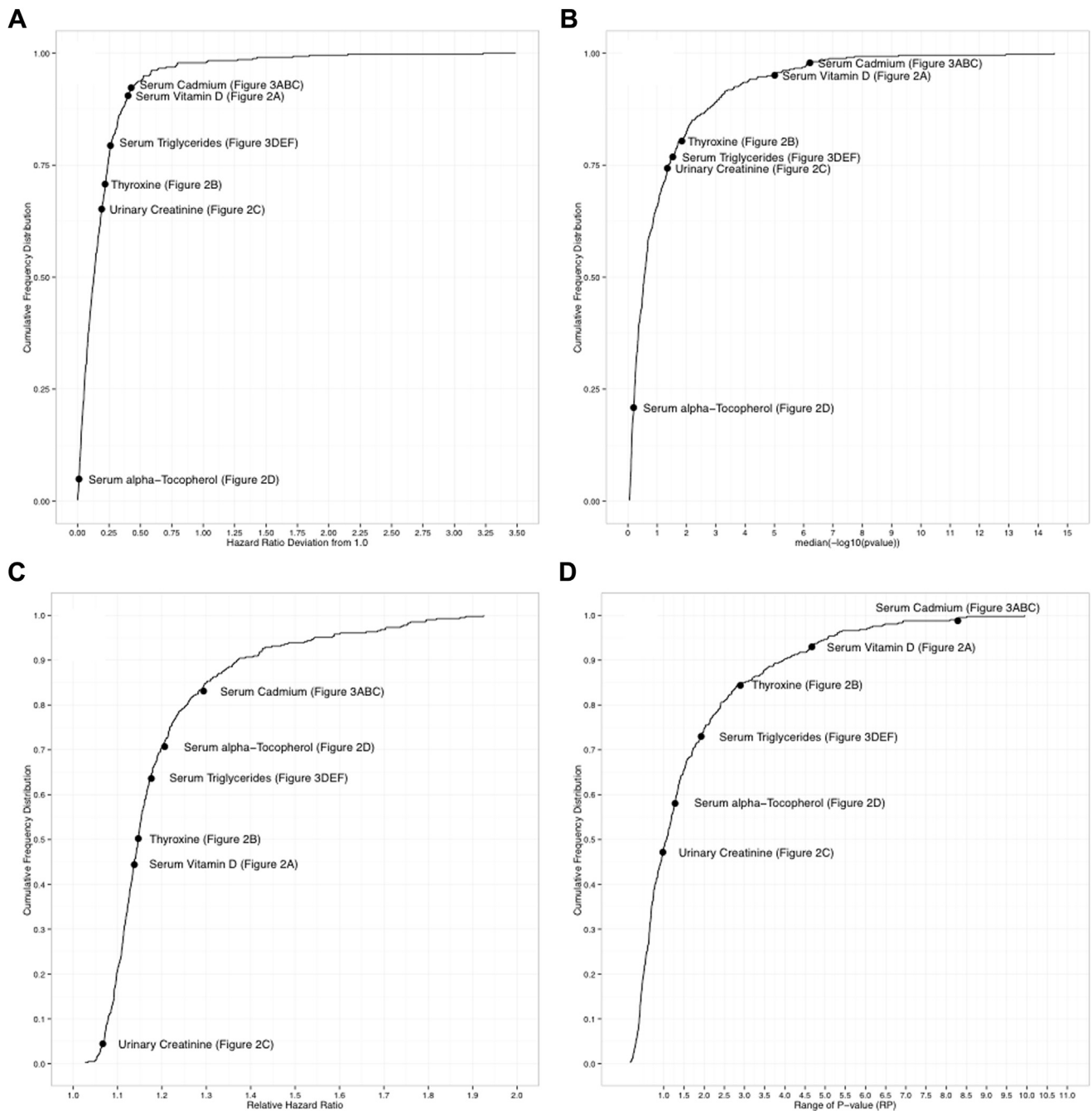


Fig. 4. Cumulative distributions of vibration of effects (VoE) for 417 variables. (A) Absolute deviation of HR from 1, (B) $\log_{10}(P\text{-value})$, (C) relative hazard ratio (RHR), (D) relative $P\text{-value}$ (RP). Examples shown in Figures 1–3 are shown in the distribution.

multimodality, such as serum cadmium and triglycerides, tended to have larger VoE. For example, serum cadmium had an RHR of 1.29 (82nd percentile) and one of the highest RPs, 8.29 (99th percentile). Serum triglycerides had an RHR of 1.18 (64th percentile) and an RP of 1.93 (73rd percentile).

4.5. Increasing the set of adjusting variables for serum vitamin D

We increased the set of adjusting variables ($n = 19$) to observe differences in the VoE distribution under new

assumptions for the association between vitamin D and mortality (see Fig. S3/Appendix at www.jclinepi.com). We observed that as the number of possible models expanded from 8,192 to 524,288, there was not a dramatic shift in the RHR/RP (see Fig. S3/Appendix at www.jclinepi.com and Fig. 2A) of 1.14/4.7 vs. 1.19/4.22 for the original VoE and the VoE under new assumptions ($n = 19$ variables), respectively. However, we did observe further attenuation of the effects as a function of number of adjusting variables and some models now indicated a null association between vitamin D and mortality (see Fig. S3/Appendix at www.jclinepi.com).

4.6. Investigation of interactions with the VoE

Effects may vary depending on different levels/categories of adjustment variables, known as “interactions.” We extended the VoE methodology to systematically consider interactions between variables of interest and adjustment variables (see Section 3). The strongest interaction included that between platelet count and total cholesterol. Specifically, we observed that the effects for platelet count were systematically opposite depending on cholesterol levels and most of the effects were most significant at higher levels of cholesterol (Fig. S4/Appendix at www.jclinepi.com). Notably, we found that VoE will increase in the presence of an interaction.

4.7. Resolution of the example on vitamin E and mortality

As shown in Figures 2D and 4D, the association between higher levels of α -tocopherol points to both higher and lower risk for mortality depending on the choice of the adjustment variables in the model in a pattern that is characteristic of what we will call the *Janus effect*, named after the two-faced Roman god. In our primer, we show that a majority (4,821 of 8,192 or 59%) of models show $HR < 1$, indicating decreased risk for death for α -tocopherol. The remainder of models indicate increased risk for death ($HR > 1$, 41%). A number of models are even significant at a nominal P -value threshold of 0.05 (81 of 8,192, or 1%). These nominally significant models ranged in their complexity, from univariate to having up to eight other adjustment variables. The association between α -tocopherol and mortality is one that is highly sensitive to model choice. This extreme VoE may explain, at least in part, the inability to replicate the observational associations in randomized trials and the hotly debated discrepancies.

5. Discussion

Almost all reported findings in observational quantitative research to date in fields such as epidemiology consider only a single or a few modeling scenarios. It is often not clear whether this or these model(s) was/were selected a priori. It is often suspected that selective reporting abounds, that is, several models are tested and only those with the most impressive results are presented with particular attraction for nominally significant results [39]. There are ongoing efforts to enhance transparency, improve reporting, and reveal whether models are prespecified, and if so, how [40]. We suggest that instead of presenting a single effect and P -value for an association of interest, one can present the median effect, the median P -value, the RHR and the RP across all possible analyses using different adjustments in addition to the pattern of the VoE, whether a Janus effect exists, and whether there are clear multimodality (and if so, due to what). Although we used here 99th and 1st percentiles to report the RHR and RP, we acknowledge

that these percentiles are arbitrary and are used for illustrative purposes. It is possible that selective reporting may cherry pick even more extreme values than these percentiles. In addition, the full number of model specifications may approach infinity. For example, if we consider not only 13 but 417 covariates, this would be 2^{417} choices. As we show here, for most associations, the magnitude of the effect and even the presence or not of nominal significance may depend on the analytical model used. Presenting the VoE metrics and plots can offer a proper picture of the stability of postulated associations. This may help avoiding surprises when some unstable associations are not replicated in subsequent studies, a common feature in observational research [9,41] that can lead to contradictory statements and endless debates.

Some of the patterns described here have been documented, albeit in different contexts. For example, Altham has described the attenuation of effect size as a function of increasing the number of covariates [42]. The opposite effect, where effect sizes increase on addition of covariates is a pattern attributed to “suppressor variables” in the psychometric literature [43]. In situations where effect sizes connote no major difference in risk for mortality, we can expect patterns such as “Janus effect.” The VoE framework allows investigators to assess empirically the relative frequency of these patterns.

Model selection may be dependent on the data set, and overfitting causes poorer performance in independent data sets [43–45]. Methods such as Bayesian model averaging (BMA) [46,47], relative importance based on variance decomposition [48], and bootstrap-based procedures aim to avoid selection of overfit models. Specifically, BMA does “all-subsets regression” to infer the best model of those tested while taking into account posterior probability of models, and it has been applied to ascertain model uncertainty [49]. Relative importance estimates the increase of variance explained when individual variables are added to a linear regression model. Stability of models, such as those ascertained via relative importance measures, can also be assessed through other simulation methods, such as the “bootstrap” [44]. In bootstrap-based procedures, an automated model selection procedure is performed on many random samples from the data set. Variables that are selected in multiple random samplings of the data are hypothesized to be robust candidates to describe associations in an independent data set. Although related to our method, these methods aim to select the best possible model(s). The VoE, on the other hand, focuses on how much the effect sizes change because of model selection. Even though a specific model may have better statistical support over other models, it is often impossible to isolate a single model that conceptually is definitely the best versus thousands of competitors.

Here, we described the VoE empirically as a function of estimates from 8,192 specifications. However, these different specifications may not define the entire scope of model flexibility that can be encountered in large discovery-based observational studies with many variables.

Visible and invisible “multiplicities” can be introduced anywhere in the design and analysis process [50,51]. For example, eligibility criteria can be altered to include fewer or more participants in the analysis; in our study, we computed the VoE over a fixed sample size. Furthermore, we do not explore flexibility in outcome definition (e.g., cardiovascular-related mortality vs. all-cause mortality). Flexibility in defining outcomes may result in outcome reporting bias with further instability in the results [52,53]. The VoE quantifies how the single or few of “main effect [s]” reported in biomedical investigations are but a handful of findings in a vast space of possible findings, and perhaps, it may behoove of investigators to report all possible “main effects” for a modeling scenario.

We emphasize there is no consensus for what variables should be included in a model or a model selection procedure even for high priority risk factors and outcomes, such as physical activity and mortality. For example, in a comprehensive meta-analysis of association between physical activity and mortality, Samitz et al. observed different combinations of 26 variables were used in individual investigations [29]. Of the 80 independent studies, the most common variables in the individual analyses included age (in 86.3% of studies), smoking (78%), and BMI (55%). But, some studies included mental disorder (5%), hormone replacement therapy (4%), liver disease (4%), and renal disease (4%). Because of lack of consensus, we selected for exploration the most common adjusting variables from these independent studies.

Nevertheless, additional and/or other adjustment variables can be chosen. In a supplementary analysis for vitamin D, we selected 19 variables resulting in exponentially more models to compute (524,288) and impractical as codified here. Estimating the VoE for 20 or more variables may be possible with algorithms such as “leaps and bounds” [54].

The fundamental analytical methods used to compute effects, such as regression, are also subject to choice. For example, we conducted survival analyses with the Cox semiparametric tool, but one could have chosen other modeling methods. An empirical study testing eight different statistical methods for adverse effect identification found that results were often discrepant [55].

The way data sets are sampled and data set heterogeneity may also lead to different results [56]. If several of these additional options are considered as mainstream options, the VoE is expected to increase further. Thus, VoE should be presented with specification of the type and range of choices that it has encompassed. Finally, in any observational research, there may be many unaccounted, unmeasured, or unknown confounders. These may invalidate a claimed association, even when it shows limited VoE and consistently strong and statistically significant results based on the models that can be estimated based from available data and measurements. For example, although serum vitamin D was found to have consistently significant results in the VoE analyses with 15 considered variables, other studies do not show such consistency

[57] and this inconsistency became apparent only when 19 variables were considered. Therefore, the VoE estimates should be seen as a lower limit of the uncertainty that is inherent in observational associations. Routine adoption of the VoE concept and analyses may place observational associations in a more proper context, acknowledging their potential instability.

The research process, including model selection, may be driven by incentives for the investigator themselves [58]. To avoid an excess of cherry picking of model specifications with most favorable/significant results, incentives for researchers must change. Journal editors, funding agencies, and referees must coordinate efforts to ensure researchers are reporting transparent and unbiased results. Furthermore, analytical software and computational infrastructure must be broadly accessible. Toward this goal, the data, code, and figures for this primer are available in a repository located at <http://chiragjpgroup.org/voe>.

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Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclinepi.2015.05.029>.

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